COMPUTER GRAPHICS · LECTURE 05

Computer Graphics — Lecture 5: Aliasing, Sampling & Reconstruction

محاضرة من كورس CMU 15-462/662 عن التعرّج (Aliasing) في الصور والصوت، عملية أخذ العينات (Sampling) وإعادة البناء (Reconstruction)، تحليل الترددات (Frequencies) في الإشارات والصور، الفلاتر (Low/High/Band-pass) والأمثلة الكلاسيكية زي wagon wheel effect و sin(x^2 + y^2).

32	12	6	50
MCQ	Essay	Computational	Total

المادة: Computer Graphics (CMU 15-462/662)

المحاضرة: Lecture 05 — Aliasing, Sampling & Reconstruction

اللغة: English (أسئلة وإجابات) + Arabic (شرح وترجمة)

التوزيع: MCQ correct = 8 لكل اختيار (A, B, C, D)

Questions & Answers سؤال 50

Q1 MCQ HIGH

1. What is the primary goal of sampling and reconstruction in signal processing?

إيه هو الهدف الأساسي من أخذ العينات وإعادة البناء في معالجة الإشارات؟

A To compress the original signal as much as possible

B To reproduce the original signal as accurately as possible

C To increase the frequency of the original signal

D To convert the signal into a lower-resolution form

✅ الإجابة الصحيحة: B — To reproduce the original signal as accurately as possible

💡 ليه صح:

الشريحة بتقول صراحة: الهدف هو إعادة إنتاج الإشارة الأصلية بأكبر دقة ممكنة (reproduce original signal as accurately as possible). الاختيارات التانية غلط لأن الضغط أو تقليل الدقة أهداف ثانوية مش الأساسية.

📍 المرجع: Lecture 5 — Sampling & Reconstruction (intro slide)

Tags: Sampling Reconstruction Goal

Q2 MCQ HIGH

2. Aliasing is best described as:

تعريف التعرّج (Aliasing) هو:

A Low frequencies in the original signal appearing as high frequencies after reconstruction

B High frequencies in the original signal masquerading as low frequencies after reconstruction due to undersampling

C The complete loss of all frequencies during sampling

D The process of converting a digital signal back to analog

✅ الإجابة الصحيحة: B — High frequencies in the original signal masquerading as low frequencies after reconstruction due to undersampling

💡 ليه صح:

التعريف موجود حرفيًا في الشريحة: الترددات العالية في الإشارة الأصلية بتتنكر في هيئة ترددات منخفضة بعد إعادة البناء بسبب أخذ عينات قليلة (undersampling). الاختيار الأول عكس الكلام، والباقي مش تعريف صحيح.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Aliasing Definitions

Q3 MCQ HIGH

3. In the sampling/reconstruction pipeline, the intermediate 'digital information' stage represents:

في خط أخذ العينات وإعادة البناء، مرحلة 'digital information' بتعمل إيه؟

A The reconstructed continuous signal

B The original continuous signal before any processing

C The discrete samples taken from the original continuous signal

D A higher-frequency version of the original signal

✅ الإجابة الصحيحة: C — The discrete samples taken from the original continuous signal

💡 ليه صح:

الخط هو continuous signal → sample → digital information → reconstruct → continuous signal. فـ digital information هي العينات المدمجة (discrete samples) المأخوذة من الإشارة الأصلية. الاختيارات التانية بتمثل مراحل تانية في الخط.

📍 المرجع: Lecture 5 — Sampling & Reconstruction pipeline diagram

Tags: Sampling Pipeline

Q4 MCQ HIGH

4. According to the lecture, a 1D signal can be expressed as:

حسب المحاضرة، الإشارة الـ 1D ممكن تتعرض كـ:

A A single constant value

B A single high-frequency wave

C A random sequence of values

D A superposition (sum) of frequencies

✅ الإجابة الصحيحة: D — A superposition (sum) of frequencies

💡 ليه صح:

الشريحة بتقول إن إشارة 1D ممكن تتعبّر عنها كـ superposition of frequencies (مجموع ترددات). ده أساس فوريه. الاختيارات التانية بتعكس سوء فهم للنظرية.

📍 المرجع: Lecture 5 — 1D signal / frequencies slide

Tags: Frequencies Fourier 1D signals

Q5 MCQ HIGH

5. In the example f(x) = f1(x) + 0.75 f2(x) + 0.5 f4(x), the three component signals were:

في المثال f(x) = f1 + 0.75 f2 + 0.5 f4، الإشارات الثلاثة f1, f2, f4 كانت:

A sin(x), sin(2x), sin(4x)

B sin(pi*x), sin(3*pi*x), sin(5*pi*x)

C cos(x), cos(2x), cos(4x)

D sin(pi*x), sin(2*pi*x), sin(4*pi*x)

✅ الإجابة الصحيحة: D — sin(pi*x), sin(2*pi*x), sin(4*pi*x)

💡 ليه صح:

الشريحة بتعرّف f1(x)=sin(pi*x)، f2(x)=sin(2*pi*x)، f4(x)=sin(4*pi*x). الاختيار التاني هو الوحيد المطابق. الاختيار الأول مفقود عامل pi، والتالت رابع بيستبدل sin بـ cos أو بيغيّر المعاملات.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Frequencies Example sin waves

Q6 MCQ HIGH

6. An audio spectrum analyzer displays:

محلل الطيف الصوتي (audio spectrum analyzer) بيعرض:

A The total energy of the signal only

B The time-domain waveform of the signal

C The amplitude of each frequency in the signal

D The number of samples taken per second

✅ الإجابة الصحيحة: C — The amplitude of each frequency in the signal

💡 ليه صح:

الشريحة بتقول: محلل الطيف الصوتي بيعرض سعة كل تردد (amplitude of each frequency). ده عكس شكل الموجة اللي هو في المجال الزمني. الاختيارات التانية مش بتاع الطيف.

📍 المرجع: Lecture 5 — Audio spectrum analyzer slide

Tags: Spectrum analyzer Frequencies

Q7 MCQ HIGH

7. In the audio spectrum analyzer shown in the lecture, low frequencies correspond to:

في محلل الطيف الصوتي، الترددات المنخفضة بتقابل:

A Treble (high-pitched sounds)

B Background noise only

C Silence

D Bass (low-pitched sounds)

✅ الإجابة الصحيحة: D — Bass (low-pitched sounds)

💡 ليه صح:

الشريحة بتقول صراحة: الترددات المنخفضة = bass (صوت جهير). ده مكتوب حرفيًا جنب المنطقة المنخفضة في الرسمة. الاختيار الأول غلط لأن treble هو الترددات العالية.

📍 المرجع: Lecture 5 — Audio spectrum analyzer slide

Tags: Spectrum analyzer | Audio | Frequencies

Q8 MCQ HIGH

8. In the lecture's audio demo, the tone frequency is given by w(t) = 6000*t. What happens as t increases?

في المثال الصوتي w(t)=6000*t، إيه اللي بيحصل لما t يزيد؟

A The frequency increases over time and eventually aliasing occurs

B The tone stays at a constant frequency

C The audio becomes silent

D The amplitude decreases over time

✅ الإجابة الصحيحة: A — The frequency increases over time and eventually aliasing occurs

💡 ليه صح:

الشريحة بتسأل 'What happens if we increase w over time?' وبعدين بتعرض تجربة بتوضح إن aliasing بيحصل. التردد بيزيد بشكل خطي وبيتعدى قدرة أخذ العينات، فيظهر التعرّج. الاختيار الأول مخالف لطبيعة w(t)=6000*t أصله.

📍 المرجع: Lecture 5 — Aliasing in Audio slide

Tags: Aliasing | Audio demo | Frequencies

Q9 MCQ HIGH

9. Aliasing occurs because:

التعرّج بيحصل بسبب:

A The signal is sampled too frequently

B High-frequency signals are insufficiently sampled (undersampled)

C The signal contains no high frequencies

D The reconstruction uses too many samples

✅ الإجابة الصحيحة: B — High-frequency signals are insufficiently sampled (undersampled)

💡 ليه صح:

الشريحة بتقول صراحة: 'Undersampling high-frequency signals results in aliasing'. أخذ عينات قليلة لإشارة عالية التردد هو السبب. الاختيار الأول عكس الصحيح — أخذ عينات كتير بيلغي التعرّج مش بيعمله.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Aliasing Undersampling

Q10 MCQ HIGH

10. After undersampling a high-frequency signal, the reconstruction appears to come from:

بعد أخذ عينات قليلة من إشارة عالية التردد، إعادة البناء بتظهر كأنها جاية من:

A A higher-frequency signal than the original

B A noise-free version of the original

C A signal with zero frequency

D A low-frequency signal

✅ الإجابة الصحيحة: D — A low-frequency signal

💡 ليه صح:

الشريحة بتقول: 'reconstruction appears to be from a low frequency signal'. ده جوهر التعرّج — الترددات العالية بتتنكر في هيئة ترددات منخفضة. الاختيار الأول غلط تمامًا لأن التعرّج بيقلل التردد الظاهري مش بيزيده.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Aliasing Reconstruction

Q11 MCQ HIGH

11. Which type of signal can be sampled adequately for accurate reconstruction (according to the lecture)?

أي نوع إشارة ممكن يتعمله أخذ عينات كافي لإعادة بناء دقيقة؟

A Only high-frequency signals

B Low-frequency signals

C Only signals with infinite frequency

D All signals regardless of frequency

✅ الإجابة الصحيحة: B — Low-frequency signals

💡 ليه صح:

الشريحة بتقول: 'Low-frequency signal: sampled adequately for accurate reconstruction'. الترددات المنخفضة بتتحتاج عينات أقل عشان تتقال صح. الترددات العالية بتحتاج عينات أكتر، ولما متوفرش، بيحصل aliasing.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Sampling Reconstruction Frequencies

Q12 MCQ HIGH

12. According to the lecture, images can also be:

حسب المحاضرة، الصور كمان ممكن:

A Decomposed into 'frequencies'

B Stored only as a single frequency

C Reconstructed without any sampling

D Compressed into a single pixel

✅ الإجابة الصحيحة: A — Decomposed into 'frequencies'

💡 ليه صح:

الشريحة بتقول حرفيًا: 'Images can also be decomposed into frequencies'. ده امتداد لنفس فكرة تحليل الإشارات للترددات، بس للمجال المكاني (spatial) بدل الزماني.

📍 المرجع: Lecture 5 — Images as frequencies slide

Tags: Images Frequencies Decomposition

Q13 MCQ HIGH

13. In image frequency analysis, low frequencies typically correspond to:

في تحليل ترددات الصور، الترددات المنخفضة بتقابل:

A Sharp edges

B High contrast regions

C Random noise

D Smooth gradients

✅ الإجابة الصحيحة: D — Smooth gradients

💡 ليه صح:

الشريحة بتقول صراحة: 'Low frequencies only (smooth gradients)'. التردد المنخفض بيمثل تغيّر بطيء في قيم البيكسل (تدرّج ناعم). التردد العالي بيمثل التغيّر السريع زي الحدود.

📍 المرجع: Lecture 5 — Low frequencies (low-pass) slide

Tags: Images Frequencies Low-pass

Q14 MCQ HIGH

14. A low-pass filter applied to an image:

الفلتر low-pass لما يتطبّق على صورة بيعمل إيه؟

A Removes low frequencies and keeps high frequencies

B Keeps all frequencies unchanged

C Keeps low frequencies and removes high frequencies

D Increases the amplitude of all frequencies

✅ الإجابة الصحيحة: C — Keeps low frequencies and removes high frequencies

💡 ليه صح:

اسم الفلتر 'low-pass' معناه يخلّي الترددات المنخفضة تعدي (pass) ويوقف العالية. بعد الفلتر: الترددات فوق cutoff بتبقى صفر. النتيجة المكانية: smooth gradients فقط.

📍 المرجع: Lecture 5 — Low frequencies / low-pass filter slide

Tags: Filters Low-pass Images

Q15 MCQ HIGH

15. After applying a low-pass filter, the spectrum shows:

بعد تطبيق فلتر low-pass، الطيف بيبين:

A All frequencies have the same magnitude

B Only high frequencies remain

C All frequencies above the cutoff have 0 magnitude

D All frequencies are amplified

✅ **الإجابة الصحيحة: C — All frequencies above the cutoff have 0 magnitude**

💡 **ليه صح:**

الشريحة بتقول: 'All frequencies above cutoff have 0 magnitude'. ده تعريف low-pass filter في مجال التردد — بيصفر الترددات فوق نقطة القطع. الاختيارات التانية مخالفة لتعريف الفلتر.

📍 المرجع: Lecture 5 — Low frequencies (after low-pass) slide

Tags: Low-pass Spectrum

Q16 MCQ HIGH

16. Applying a high-pass filter to an image produces a result that contains mainly:

تطبيق فلتر high-pass على صورة بيطلّع نتيجة فيها أساسًا:

A The strongest edges

B Smooth gradients only

C A completely black image

D A blurred version of the image

✅ **الإجابة الصحيحة: A — The strongest edges**

💡 **ليه صح:**

الشريحة بتقول: 'High frequencies (edges) — Spatial domain result (strongest edges)'. الفلتر بيشيل الترددات المنخفضة (التدرّجات) وبيسيب العالية (الحدود). عشان كده النتيجة بتظهر الحدود بوضوح.

📍 المرجع: Lecture 5 — High frequencies (high-pass) slide

Tags: High-pass Edges Filters

Q17 MCQ HIGH

17. After applying a high-pass filter, the spectrum shows:

بعد تطبيق فلتر high-pass، الطيف بيبين:

A All frequencies above threshold have 0 magnitude

B Only the DC component remains

C All frequencies are kept unchanged

D All frequencies below threshold have 0 magnitude

✅ **الإجابة الصحيحة: D — All frequencies below threshold have 0 magnitude**

💡 **ليه صح:**

الشريحة بتقول صراحة: 'All frequencies below threshold have 0 magnitude'. ده عكس الـ low-pass — الـ high-pass بيشيل الترددات المنخفضة (تحت العتبة) وبيسيب العالية.

📍 المرجع: Lecture 5 — High frequencies (high-pass) slide

Tags: High-pass Spectrum

Q18 MCQ HIGH

18. In image frequency analysis, high frequencies typically correspond to:

في تحليل ترددات الصور، الترددات العالية بتقابل:

A Smooth color regions

B The background of the image

C The average brightness of the image

D Edges

✅ **الإجابة الصحيحة: D — Edges**

💡 **ليه صح:**

الشريحة بتقول: 'High frequencies (edges)'. التردد العالي بيمثل تغيّر سريع في قيم البيكسل زي اللي بيحصل عند الحدود. التدرّج الناعم هو التردد المنخفض.

📍 المرجع: Lecture 5 — High frequencies (high-pass) slide

Tags: Images Frequencies High-pass

Q19 MCQ HIGH

19. A band-pass filter on an image keeps:

الفلتر band-pass على صورة بيخلّي:

A Only the lowest frequencies

B Only the highest frequencies

C Only mid-range frequencies

D All frequencies except mid-range ones

✅ **الإجابة الصحيحة: C — Only mid-range frequencies**

💡 **ليه صح:**

الشريحة بتعرض 'Mid-range frequencies — Spectrum (after band-pass filter)'. الـ band-pass بيخلّي ترددات في نطاق وسط فقط — لا العالية ولا المنخفضة. ده عكس low-pass و high-pass.

📍 المرجع: Lecture 5 — Mid-range frequencies (band-pass) slide

Tags: Band-pass Filters Mid-range

Q20 MCQ HIGH

20. According to the lecture, an image can be viewed as:

حسب المحاضرة، الصورة ممكن تتعامل معاها كـ:

A A single frequency component

B A sum of its frequency components

C A random distribution of pixels

D An array of low frequencies only

✅ **الإجابة الصحيحة: B — A sum of its frequency components**

💡 **ليه صح:**

الشريحة بتقول: 'An image as a sum of its frequency components'. ده نفس مبدأ superposition للإشارات 1D، بس مطبّق على المجال المكاني 2D.

📍 المرجع: Lecture 5 — Image as a sum of frequency components slide

Tags: Images Frequencies Decomposition

Q21 MCQ HIGH

21. The function sin(x^2 + y^2) shown in the lecture is used to demonstrate:

الدالة sin(x^2 + y^2) اللي ظهرت في المحاضرة بتوضّح:

A Temporal aliasing in audio

B The effect of low-pass filtering

C Spatial aliasing in images

D The wagon wheel effect

✅ الإجابة الصحيحة: C — Spatial aliasing in images

💡 ليه صح:

عنوان الشريحة بيقول 'Spatial aliasing: the function sin(x^2 + y^2)'. الدالة دي بتعمل حلقات (rings) متركّزة، ولما التردد بيزيد بسرعة، بيحصل aliasing مكاني — الحلقات اللي هي عالية التردد بتظهر كأنها منخفضة التردد.

📍 المرجع: Lecture 5 — Spatial aliasing: sin(x^2 + y^2) slide

Tags: Spatial aliasing sin(x^2+y^2) Examples

Q22 MCQ HIGH

22. In the sin(x^2 + y^2) example, the rings on the right side of the image illustrate:

في مثال sin(x^2 + y^2)، الحلقات على يمين الصورة بتوضّح:

A True low-frequency oscillation of the signal

B The result of applying a low-pass filter

C Aliasing from undersampling a high-frequency oscillation

D A region with no signal

✅ الإجابة الصحيحة: C — Aliasing from undersampling a high-frequency oscillation

💡 ليه صح:

الشريحة بتقول: 'Right: aliasing from undersampling high frequency oscillation makes it appear that rings are low-frequency (they are not!)'. الإشارة الحقيقية عالية التردد، لكن بسبب أخذ عينات قليلة بتظهر كأنها منخفضة التردد.

📍 المرجع: Lecture 5 — Spatial aliasing: sin(x^2 + y^2) slide

Tags: Spatial aliasing sin(x^2+y^2)

Q23 MCQ HIGH

23. In the sin(x^2 + y^2) image, the 'middle' rings represent:

في صورة sin(x^2 + y^2)، الحلقات 'الوسطى' بتمثل:

A Frequencies below the cutoff

B Frequencies approaching the limit of what individual pixels can represent

C The DC component of the image

D The result of band-pass filtering

✅ **الإجابة الصحيحة: B — Frequencies approaching the limit of what individual pixels can represent**

💡 **ليه صح:**

الشريحة بتقول: 'Middle: ring frequency approaches limit of what we can represent w/ individual pixels'. هنا التردد بيقرب من حد قدرة البيكسل الفردي على تمثيله، وده بيدّي إنذار قبل ما aliasing يبدأ على اليمين.

📍 المرجع: Lecture 5 — Spatial aliasing: sin(x^2 + y^2) slide

Tags: Spatial aliasing | sin(x^2+y^2) | Pixel resolution

Q24 MCQ HIGH

24. The wagon wheel effect is a classic example of:

ظاهرة wagon wheel effect هي مثال كلاسيكي على:

A Spatial aliasing

B Color aliasing

C Audio aliasing

D Temporal aliasing

✅ **الإجابة الصحيحة: D — Temporal aliasing**

💡 **ليه صح:**

عنوان الشريحة بيقول: 'Temporal aliasing: wagon wheel effect'. 'Temporal' معناها زمني، والإشارة هنا متغيّرة في الزمن (عجلة بتلف)، فالتعرّج بيحصل في المحور الزمني مش المكاني.

📍 المرجع: Lecture 5 — Temporal aliasing: wagon wheel effect slide

Tags: Temporal aliasing | Wagon wheel

Q25 MCQ HIGH

25. According to the lecture, the wagon wheel effect occurs because:

حسب المحاضرة، wagon wheel effect بيحصل لأن:

- **A** The camera's frame rate is too low for the rapidly spinning wheel
- **B** The wheel is not actually moving
- **C** The lighting is too bright
- **D** The wheel is made of reflective material

✅ **الإجابة الصحيحة: A — The camera's frame rate is too low for the rapidly spinning wheel**

💡 **ليه صح:**

الشريحة بتقول: 'Camera's frame rate (temporal sampling rate) is too low for rapidly spinning wheel'. معدّل إطارات الكاميرا (= معدّل أخذ العينات الزمني) أقل من اللازم لالتقاط سرعة العجلة، فالتعرّج الزمني بيخلّيها تبان واقفة أو بتلف في الاتجاه المعاكس.

📍 المرجع: Lecture 5 — Temporal aliasing: wagon wheel effect slide

Tags: Temporal aliasing | Wagon wheel | Frame rate

Q26 MCQ HIGH

26. The fundamental difference between spatial aliasing and temporal aliasing is:

الفرق الأساسي بين التعرّج المكاني والتعرّج الزمني هو:

- **A** Spatial occurs in images (2D space); temporal occurs in time (video frames)
- **B** Spatial is for audio; temporal is for images
- **C** Spatial happens only with color; temporal only with grayscale
- **D** There is no real difference between them

✅ **الإجابة الصحيحة: A — Spatial occurs in images (2D space); temporal occurs in time (video frames)**

💡 **ليه صح:**

التعرّج المكاني بيحصل في المجال المكاني (صور 2D زي sin(x^2+y^2))، والتعرّج الزمني بيحصل في المجال الزمني (فيديو، زي wagon wheel). السبب في الاتنين هو نفسه (undersampling)، بس المحور مختلف.

📍 المرجع: Lecture 5 — Spatial aliasing vs Temporal aliasing slides

Tags: Spatial aliasing | Temporal aliasing | Comparison

Q27 MCQ HIGH

27. In the f(x) = f1(x) + 0.75 f2(x) + 0.5 f4(x) example, the largest amplitude coefficient belongs to:

في مثال f(x)=f1+0.75 f2+0.5 f4، أكبر معامل سعة موجود في:

A f1 (coefficient = 1)

B f2 (coefficient = 0.75)

C f4 (coefficient = 0.5)

D All coefficients are equal

✅ الإجابة الصحيحة: A — f1 (coefficient = 1)

💡 ليه صح:

المعاملات: f1 بـ 1، f2 بـ 0.75، f4 بـ 0.5. أكبر معامل هو 1 الخاص بـ f1. ده بيمثل إن التردد الأساسي (الأدنى) ليه أعلى مساهمة في الإشارة المركّبة.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Frequencies Example Amplitude

Q28 MCQ MEDIUM

28. The wavelength/period of f1(x) = sin(pi*x) compared to f4(x) = sin(4*pi*x) means:

مقارنة f1=sin(pi*x) بـ f4=sin(4*pi*x) معناه:

A f1 has the shortest wavelength (highest frequency)

B Both have the same frequency

C f4 has the highest frequency of the three

D f1 has zero frequency

✅ الإجابة الصحيحة: C — f4 has the highest frequency of the three

💡 ليه صح:

معامل التردد في f4 هو 4*pi، أكبر من 2*pi في f2 و pi في f1. فـ f4 عندها أعلى تردد بين الإشارات الثلاثة. ده بيربط فكرة إن المعامل الأكبر = تردد أعلى.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Frequencies sin waves Comparison

Q29 MCQ MEDIUM

29. The 'sample' stage in the sampling/reconstruction pipeline converts:

مرحلة 'sample' في خط المعالجة بتحوّل:

- **A** Continuous signal to discrete samples (digital information)
- **B** Digital information back to continuous signal
- **C** Low frequencies to high frequencies
- **D** Audio signal to image signal

✅ **الإجابة الصحيحة: A** — Continuous signal to discrete samples (digital information)

💡 **ليه صح:**

الترتيب هو continuous → sample → digital → reconstruct. فـ sample بتحوّل الإشارة المستمرة لعينات منفصلة (digital). الـ reconstruct هي اللي بتعمل العكس — بتحوّل digital لـ continuous.

📍 المرجع: Lecture 5 — Sampling & Reconstruction pipeline diagram

Tags: Sampling Pipeline Digital

Q30 MCQ MEDIUM

30. The 'reconstruct' stage in the pipeline is responsible for:

مرحلة 'reconstruct' في خط المعالجة مسؤولة عن:

- **A** Converting digital information back to an approximate continuous signal
- **B** Sampling the continuous signal
- **C** Increasing the frequency of the signal
- **D** Adding aliasing to the signal

✅ **الإجابة الصحيحة: A** — Converting digital information back to an approximate continuous signal

💡 **ليه صح:**

الـ reconstruct بياخد digital information (عينات) ويعمل منها إشارة مستمرة تقريبية. الهدف مكتوب في الشريحة: reproduce original signal as accurately as possible. لو أخد العينات مش كفاية، الـ reconstruct بتعمل aliasing.

📍 المرجع: Lecture 5 — Sampling & Reconstruction pipeline diagram

Tags: Reconstruction Pipeline

Q31 MCQ MEDIUM

31. In the spatial aliasing image of sin(x^2 + y^2), the rings on the 'center-left' represent:

في صورة sin(x^2+y^2)، الحلقات في 'الوسط-يسار' بتمثل:

A Aliasing artifacts

B The actual (true) low-frequency oscillation of the signal

C The result of a high-pass filter

D Pixels outside the image bounds

✅ الإجابة الصحيحة: B — The actual (true) low-frequency oscillation of the signal

💡 ليه صح:

الشريحة بتقول: 'Rings in center-left: Actual signal (low frequency oscillation)'. هنا الإشارة لسه منخفضة التردد بشكل حقيقي وبتتمثّل صح. الـ aliasing بيبدأ على اليمين لما التردد يزيد.

📍 المرجع: Lecture 5 — Spatial aliasing: sin(x^2 + y^2) slide

Tags: Spatial aliasing sin(x^2+y^2)

Q32 MCQ MEDIUM

32. After reconstruction, an undersampled high-frequency signal appears:

بعد إعادة البناء، إشارة عالية التردد اتاخد منها عينات قليلة بتظهر:

A Identical to the original

B As if it came from a low-frequency signal

C As a constant DC value

D As a higher-frequency signal than before

✅ الإجابة الصحيحة: B — As if it came from a low-frequency signal

💡 ليه صح:

الشريحة بتقول صراحة: 'reconstruction appears to be from a low frequency signal'. التردد العالي بيتنكر في هيئة تردد منخفض — ده التعريف الأساسي للـ aliasing. الاختيار الأول غلط ولو كان صح مكانش فيه مشكلة.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Aliasing Reconstruction Frequencies

E1 ESSAY

1. Define aliasing and explain why it occurs.

عرّف التعرّج (Aliasing) واشرح ليه بيحصل.

✅ **الإجابة النموذجية:**

Aliasing is the phenomenon where high frequencies in the original signal masquerade as low frequencies after reconstruction, due to undersampling.

It occurs because the sampling rate is too low to capture the rapid changes of the high-frequency components. When the signal is reconstructed from these insufficient samples, the high-frequency content folds back into the low-frequency range, producing an apparent low-frequency signal that does not exist in the original.

📍 المرجع: Lecture 5 — Undersampling / Aliasing slide

Tags: Aliasing Definitions

الأولوية: high

E2 ESSAY

2. Explain the sampling and reconstruction pipeline, listing its main stages.

اشرح خط أخذ العينات وإعادة البناء واذكر مراحله الأساسية.

✅ **الإجابة النموذجية:**

The pipeline consists of four stages:

1. Continuous signal (original): the true analog signal we want to capture.

2. Sample: discretize the continuous signal into a finite set of values.

3. Digital information: the discrete samples stored/processed as digital data.

4. Reconstruct: convert the digital samples back into an approximate continuous signal.

The goal of the whole pipeline is to reproduce the original signal as accurately as possible. If the sampling rate is insufficient for the signal's high frequencies, aliasing occurs at the reconstruction stage.

📍 المرجع: Lecture 5 — Sampling & Reconstruction pipeline diagram

Tags: Sampling Reconstruction Pipeline

الأولوية: high

E3 ESSAY

3. Explain how a 1D signal can be expressed as a superposition of frequencies, using the lecture's example.

اشرح إزاي إشارة 1D ممكن تتعبر كـ superposition للترددات، باستخدام مثال المحاضرة.

✅ **الإجابة النموذجية:**

A 1D signal can be expressed as a sum (superposition) of multiple sinusoidal waves, each at a different frequency and amplitude.

In the lecture example:

f1(x) = sin(pi*x)

f2(x) = sin(2*pi*x)

f4(x) = sin(4*pi*x)

f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x)

Here f1 is the lowest frequency, f4 is the highest, and each is weighted by an amplitude coefficient (1, 0.75, 0.5). Adding these three sinusoids produces a richer, more complex signal. This is the foundation of frequency-domain analysis (Fourier decomposition).

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Frequencies Fourier 1D signals

الأولوية: high

E4 ESSAY

4. Compare spatial aliasing and temporal aliasing. Give one example of each from the lecture.

قارن بين التعرّج المكاني (Spatial) والتعرّج الزمني (Temporal)، واذكر مثال على كل واحد من المحاضرة.

✅ الإجابة النموذجية:

Spatial aliasing:

- Occurs in the spatial domain (images, 2D).

- Cause: insufficient spatial sampling (pixel resolution) for high-frequency spatial oscillations.

- Example: the function sin(x^2 + y^2) shows rings; on the right side of the image, high-frequency oscillations appear as low-frequency rings due to undersampling by individual pixels.

Temporal aliasing:

- Occurs in the time domain (video, animation).

- Cause: camera frame rate (temporal sampling rate) is too low for a rapidly moving/spinning object.

- Example: the wagon wheel effect — a spinning wheel appears to rotate backwards, stand still, or move slower than its true rotation because the camera cannot sample its motion fast enough.

Both share the same underlying cause (undersampling) but differ in the axis along which sampling is insufficient.

📍 المرجع: Lecture 5 — Spatial aliasing + Temporal aliasing slides

Tags: Spatial aliasing | Temporal aliasing | Comparison

الأولوية: high

E5 ESSAY

5. Explain what low-pass, band-pass, and high-pass filters do to an image, and describe the spatial-domain result of each.

اشرح إيه اللي بيعمله كل من فلتر low-pass و band-pass و high-pass على صورة، ووصف النتيجة في المجال المكاني لكل فيهم.

✅ **الإجابة النموذجية:**

Low-pass filter:

- Keeps low frequencies; sets all frequencies above the cutoff to zero magnitude.

- Spatial result: smooth gradients only — edges are blurred out.

Band-pass filter:

- Keeps only mid-range frequencies; removes both the lowest and highest components.

- Spatial result: a mix that contains neither the smoothest gradients nor the sharpest edges — only mid-detail structure.

High-pass filter:

- Keeps high frequencies; sets all frequencies below a threshold to zero magnitude.

- Spatial result: the strongest edges of the image are emphasized, while smooth regions are suppressed.

📍 المرجع: Lecture 5 — Low / Mid-range / High frequencies slides

Tags: Low-pass Band-pass High-pass Filters

الأولوية: high

E6 ESSAY

6. Explain why images can be decomposed into frequencies, and state what low and high frequencies correspond to in images.

اشرح ليه الصور ممكن تتفكّك لترددات، واذكر الترددات المنخفضة والعالية بتقابل إيه في الصور.

✅ **الإجابة النموذجية:**

Just like a 1D audio signal can be expressed as a superposition of sinusoidal frequencies, a 2D image can also be decomposed into 'frequencies' — in this case, spatial frequencies that describe how rapidly pixel intensities change across the image.

Low frequencies correspond to smooth gradients — regions where pixel values change slowly, such as flat color areas or gradual shading.

High frequencies correspond to edges — regions where pixel values change sharply, such as object boundaries and fine detail.

An image can therefore be viewed as a sum of its frequency components, and filtering operations (low-pass, high-pass, band-pass) selectively keep or remove these components.

📍 المرجع: Lecture 5 — Images as frequencies / Low / High slides

Tags: Images Frequencies Decomposition

الأولوية: high

E7 ESSAY

7. Explain the wagon wheel effect and its cause.

اشرح ظاهرة wagon wheel effect وسببها.

✅ **الإجابة النموذجية:**

The wagon wheel effect is a form of temporal aliasing in which a spinning wheel appears to rotate more slowly than its true rotation, appears stationary, or even appears to rotate in the opposite direction.

Cause: the camera's frame rate (the temporal sampling rate) is too low for the rapidly spinning wheel. When samples are taken too infrequently relative to the wheel's rotation speed, the reconstructed motion is ambiguous — the high-frequency rotation 'masquerades' as a lower-frequency (or even reversed) rotation.

This is the temporal counterpart of spatial aliasing: the same underlying principle (undersampling a high-frequency signal) but applied along the time axis instead of the spatial axis.

📍 المرجع: Lecture 5 — Temporal aliasing: wagon wheel effect slide

Tags: Temporal aliasing Wagon wheel

الأولوية: high

E8 ESSAY

8. Explain the sin(x^2 + y^2) example and what it demonstrates about spatial aliasing. Describe the three regions visible in the image.

اشرح مثال sin(x^2+y^2) وإيه اللي بيوضّحه عن التعرّج المكاني. وصف المناطق الثلاثة المرئية في الصورة.

✅ **الإجابة النموذجية:**

The function sin(x^2 + y^2) produces concentric rings centered at the origin, and as we move away from the origin the ring frequency increases.

Three regions are visible in the rendered image:

1. Center-left (rings in the actual signal): the true low-frequency oscillation of the signal — these rings are genuinely low-frequency and are correctly represented.

2. Middle: the ring frequency approaches the limit of what individual pixels can represent. This is the boundary where the sampling resolution becomes insufficient.

3. Right: aliasing from undersampling the high-frequency oscillation. The actual signal is high-frequency here, but due to insufficient pixel resolution, it appears as low-frequency rings.

The example demonstrates that spatial aliasing makes high-frequency spatial content masquerade as low-frequency content — the same defining property of aliasing, but in the spatial (image) domain.

📍 المرجع: Lecture 5 — Spatial aliasing: sin(x^2 + y^2) slide

Tags: Spatial aliasing sin(x^2+y^2) Examples

الأولوية: high

E9 ESSAY

9. State the goal of sampling and reconstruction, and explain how aliasing violates this goal.

اذكر هدف أخذ العينات وإعادة البناء، واشرح إزاي aliasing بيخالف الهدف ده.

✅ **الإجابة النموذجية:**

The goal of sampling and reconstruction is to reproduce the original continuous signal as accurately as possible.

Aliasing violates this goal because, when the sampling rate is too low for the signal's high-frequency content, the reconstructed signal no longer matches the original. Instead, high frequencies in the original signal masquerade as low frequencies in the reconstruction. The reconstructed signal appears to come from a low-frequency signal that was never in the original — the output is a 'false identity' (an alias) of the input.

Therefore, aliasing directly breaks the 'reproduce as accurately as possible' objective: the reconstruction is not a faithful copy of the original.

📍 المرجع: Lecture 5 — Sampling & Reconstruction goal slide

Tags: Sampling Reconstruction Aliasing Goal

الأولوية: high

E10 ESSAY

10. Explain the audio aliasing demo in which the tone frequency is w(t) = 6000*t. What happens and why?

اشرح تجربة التعرّج الصوتي اللي فيها تردد النغمة w(t)=6000*t. إيه اللي بيحصل وليه؟

✅ **الإجابة النموذجية:**

In the demo, the angular frequency of the tone is set to w(t) = 6000*t, meaning the frequency increases linearly over time.

At the beginning (small t), the frequency is low enough that the sampling rate captures it accurately, and the tone plays as expected.

As t increases, the frequency eventually exceeds what the (fixed) sampling rate can represent. Beyond this point, the high-frequency tone cannot be reconstructed faithfully — instead, it is misinterpreted as a lower-frequency tone due to aliasing.

The result is that the tone no longer sounds like a steadily rising pitch; instead, unexpected lower-frequency artifacts appear. This illustrates the general principle that increasing the signal's frequency beyond the capacity of the sampling rate produces aliasing.

📍 المرجع: Lecture 5 — Aliasing in Audio slide

Tags: Aliasing Audio demo Frequencies

الأولوية: high

E11 ESSAY

11. Explain what an audio spectrum analyzer shows and how it relates to the idea of decomposing a signal into frequencies.

اشرح إيه اللي بيعرضه محلل الطيف الصوتي وإزاي بيرتبط بفكرة تفكيك الإشارة لترددات.

✅ **الإجابة النموذجية:**

An audio spectrum analyzer shows the amplitude of each frequency present in the signal.

Instead of displaying the signal as a time-domain waveform (amplitude versus time), it displays a frequency-domain representation: each position along the frequency axis corresponds to a specific frequency, and the height at that position represents how much of that frequency is present.

In the lecture's example, the low-frequency region corresponds to bass, and the high-frequency region corresponds to treble. This visualization directly illustrates the principle that a 1D signal can be expressed as a superposition (sum) of frequencies, each with its own amplitude — the spectrum analyzer simply makes this decomposition visible.

📍 المرجع: Lecture 5 — Audio spectrum analyzer slide

Tags: Spectrum analyzer Frequencies Audio

الأولوية: medium

E12 ESSAY

12. List the three frequency ranges produced by the three image filters shown in the lecture (low-pass, band-pass, high-pass), and state which spatial feature each emphasizes.

اذكر النطاقات الترددية الثلاثة اللي بيطلّعها كل من الفلاتر الثلاثة، واذكر الخاصية المكانية اللي بيركّز عليها كل فيهم.

✅ **الإجابة النموذجية:**

1. Low-pass filter:

- Frequency range kept: low frequencies only (all frequencies above the cutoff are set to zero magnitude).

- Spatial feature emphasized: smooth gradients.

2. Band-pass filter:

- Frequency range kept: mid-range frequencies only (both low and high are removed).

- Spatial feature emphasized: medium-scale detail — neither the smoothest gradients nor the sharpest edges.

3. High-pass filter:

- Frequency range kept: high frequencies only (all frequencies below a threshold are set to zero magnitude).

- Spatial feature emphasized: the strongest edges.

📍 المرجع: Lecture 5 — Low / Mid-range / High frequencies slides

Tags: Filters Low-pass Band-pass High-pass

الأولوية: medium

E13 ESSAY

13. (Computational) Given f1(x) = sin(pi*x), f2(x) = sin(2*pi*x), f4(x) = sin(4*pi*x), and f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), compute f(0.5). Show your steps.

(مسألة حسابية) بإعطاء f1=sin(pi*x)، f2=sin(2*pi*x)، f4=sin(4*pi*x) و f(x)=f1+0.75*f2+0.5*f4، احسب f(0.5). اعرض خطواتك.

✅ الإجابة النموذجية:

Given: f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), with x = 0.5.

Step 1 — f1(0.5) = sin(pi*0.5) = sin(pi/2) = 1.

Step 2 — f2(0.5) = sin(2*pi*0.5) = sin(pi) = 0.

Step 3 — f4(0.5) = sin(4*pi*0.5) = sin(2*pi) = 0.

Step 4 — Combine: f(0.5) = 1 + 0.75*0 + 0.5*0 = 1 + 0 + 0 = 1.

Final answer: f(0.5) = 1.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Computation sin waves Example

الأولوية: high

E14 ESSAY

14. (Computational) Using the same definitions f1(x) = sin(pi*x), f2(x) = sin(2*pi*x), f4(x) = sin(4*pi*x) and f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), compute f(0.25). Use the fact that sin(pi/4) = sqrt(2)/2 (approximately 0.7071).

(مسألة حسابية) بنفس التعريفات، احسب f(0.25) مع استخدام sin(pi/4)=sqrt(2)/2 تقريبًا 0.7071.

✅ الإجابة النموذجية:

Given: f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), with x = 0.25.

Step 1 — f1(0.25) = sin(pi*0.25) = sin(pi/4) = sqrt(2)/2 ≈ 0.7071.

Step 2 — f2(0.25) = sin(2*pi*0.25) = sin(pi/2) = 1.

Step 3 — f4(0.25) = sin(4*pi*0.25) = sin(pi) = 0.

Step 4 — Combine: f(0.25) = 0.7071 + 0.75*1 + 0.5*0 = 0.7071 + 0.75 + 0 = 1.4571.

Final answer: f(0.25) ≈ 1.4571.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Computation sin waves Example

الأولوية: medium

E15 ESSAY

15. (Computational) In the audio aliasing demo, the tone's angular frequency is w(t) = 6000*t. Compute w at t = 1 second and at t = 2 seconds. Then state at which of these two times aliasing is more likely to occur, and why.

(مسألة حسابية) في تجربة التعرّج الصوتي، التردد الزاوي w(t)=6000*t. احسب w عند t=1 ثانية و t=2 ثانية. وبعدين قول في أنهي وقت من الاثنين aliasing أرجح، وليه.

✅ **الإجابة النموذجية:**

Given: w(t) = 6000*t.

Step 1 — w(1) = 6000 * 1 = 6000.

Step 2 — w(2) = 6000 * 2 = 12000.

Step 3 — Aliasing occurs when the signal's frequency is too high for the sampling rate to capture accurately. As t increases, w increases linearly, so the signal's frequency keeps rising while the sampling rate stays fixed.

Step 4 — Therefore, at t = 2 (w = 12000) aliasing is more likely than at t = 1 (w = 6000), because the higher frequency is more likely to exceed what the fixed sampling rate can represent.

Final answers: w(1) = 6000, w(2) = 12000, aliasing is more likely at t = 2.

📍 المرجع: Lecture 5 — Aliasing in Audio slide

Tags: Computation Audio demo Aliasing

الأولوية: high

E16 ESSAY

16. (Computational) Among the three component signals f1(x) = sin(pi*x), f2(x) = sin(2*pi*x), and f4(x) = sin(4*pi*x), identify which one has the highest frequency and which one has the lowest frequency. Justify your answer.

(مسألة حسابية) من بين الإشارات الثلاثة f1=sin(pi*x)، f2=sin(2*pi*x)، f4=sin(4*pi*x)، حدّد أنهي واحد عنده أعلى تردد وأنهي أدنى تردد، مع التبرير.

✅ **الإجابة النموذجية:**

Step 1 — Recall the general form sin(k*pi*x): the coefficient k determines the frequency. A larger k means more oscillations per unit of x, i.e. a higher frequency.

Step 2 — For f1, k = 1 → lowest frequency of the three.

Step 3 — For f2, k = 2 → middle frequency.

Step 4 — For f4, k = 4 → highest frequency of the three.

Final answers: f4 has the highest frequency; f1 has the lowest frequency. This is also why, in the original example, the smaller coefficient 0.5 was placed on f4 — the high-frequency component contributes less to the overall shape of the combined signal f(x).

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Computation Frequencies sin waves

الأولوية: medium

E17 ESSAY

17. (Computational) In f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), list the amplitude coefficient of each component and identify which component contributes the smallest amplitude. Explain what this means physically.

(مسألة حسابية) في f(x)=f1+0.75 f2+0.5 f4، اذكر معامل سعة كل مكوّن وحدّد أنهي فيهم بيساهم بأصغر سعة. واشرح معنى ذلك فيزيائيًا.

✅ **الإجابة النموذجية:**

Step 1 — Coefficient of f1 = 1.

Step 2 — Coefficient of f2 = 0.75.

Step 3 — Coefficient of f4 = 0.5.

Step 4 — The smallest coefficient is 0.5, which belongs to f4 (the highest-frequency component).

Physical meaning: the highest-frequency component contributes the least to the combined signal. This is consistent with the general observation that in many natural signals, the higher the frequency, the smaller its amplitude — most of the signal's energy is concentrated in the low frequencies. In image terms, this corresponds to smooth regions dominating the picture while sharp edges (high frequencies) contribute a smaller portion of the overall content.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Computation Amplitude Frequencies

الأولوية: medium

E18 ESSAY

18. (Computational) Using f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x) where f1(x)=sin(pi*x), f2(x)=sin(2*pi*x), f4(x)=sin(4*pi*x), compute f(0). Briefly explain the result.

(مسألة حسابية) بنفس التعريفات، احسب f(0) واشرح النتيجة باختصار.

✅ **الإجابة النموذجية:**

Given: f(x) = f1(x) + 0.75*f2(x) + 0.5*f4(x), with x = 0.

Step 1 — f1(0) = sin(pi*0) = sin(0) = 0.

Step 2 — f2(0) = sin(2*pi*0) = sin(0) = 0.

Step 3 — f4(0) = sin(4*pi*0) = sin(0) = 0.

Step 4 — Combine: f(0) = 0 + 0.75*0 + 0.5*0 = 0.

Final answer: f(0) = 0.

Explanation: every sine wave passes through zero at x = 0, so any linear combination of sine waves also passes through zero at x = 0. This is a property of the choice of sin (rather than cos) basis — at the origin, all sine components vanish.

📍 المرجع: Lecture 5 — 1D signal decomposition slide

Tags: Computation sin waves

الأولوية: medium

متوازن لتفادي ثبات A/B/C/D التوزيع بين RTL. الشرح والترجمة بالعربي LTR، **ملاحظة:** الأسئلة والإجابات النموذجية بالإنجليزي من كورس CMU 15-462/662 (Aliasing, Sampling & Reconstruction) الإجابة في مكان واحد. كل سؤال من محتوى المحاضرة 5.